

4 (short)

A steel strut is idealised in Fig. 4(a). The column is a prismatic hollow rectangular section fabricated from 5 mm thick steel plate and has the cross-section dimensions shown in Fig. 4(b). The steel has a Young's modulus of 210 GPa and a yield strength of 275 MPa. The strut is subject to a vertical load P applied concentric to the strut. If the failure of the strut can be assumed to be governed either by crushing or by Euler buckling, calculate the maximum load P that the strut can support. [10]

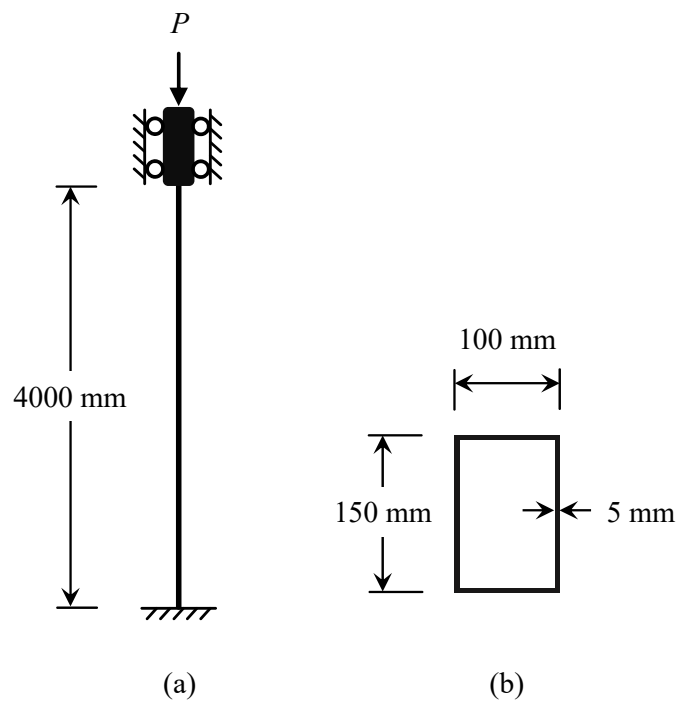


Fig. 4